Output :

ARP :

Sud : nmap -AR -sn

Starting Nmap

Nmap Scan report from 10.10.210.1

Host is up

Nmap done

Icmp :

Amap -ps -Sn 10.10.210.0/24

Starting Nmap

Nmap Scan report for 10.10.210.6

Host is up

Nmap. done

TCP :

nmap -ps 230 -Sn 10.10.210.0/24

Sudo nmap PA 80.Sn. 10.10.210/24

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NMAP TO DISCOVER LIVE HOST USING NMAP SCANS

Aim :

To study the working of the Nmap tool and discover live host in a network using different host discovery techniques such as ARP Scanning.

Steps :

1) Open the TryHackMe platform and select the Nmap room.

2) Study the network topology and identify the Source and destination hosts.

3) Use ARP request to discover the MAC address.

Host Discovery Techniques :

i) ARP Scan :

* Uses ARP requests to devices on the same Subnet.

* Hosts responding with ARP replies are considered active

ii) Host TCP / UDP Scan :

* TCP SYN ping (Sends SYN packets to selected ports

* UDP ping checks host availability through ICMP port unreachable responses

observations :

ARP Discovery :

* Active hosts respond with ARP repli
* Number of devices discovered using ARP

ICMP Discovery :

* ping requests help identify active hosts.
* ICMP scans may fail if firewall block ICMP traffic.

TCP/UDP Discovery :

* TCP SYN ping doesnot require a privileged account.
* TCP ACK ping requires a privileged account.

Result :

Thus live hosts in the network were successfully using different Nmap scanning techniques. ARP scanning was found to be effective for finding, while ICMP and UDP scanning techniques.